

Francisco Togni

Electrical Engineer | FPGA · DSP · Embedded R&D

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Summary

Electrical Engineer specialized in **Industrial Electronic Systems**, focused on FPGA, DSP, and embedded systems for real-time applications. Experience in the design and validation of low-latency digital and analog processing architectures, industrial instrumentation, and electronic control systems. Strong background in bare-metal firmware, fixed-point DSP, and hardware/software integration. C, Verilog, HDL/RTL, ARM Cortex-M, TI C2000, Xilinx. Advanced English (C1).

Experience

- **Research Collaborator**, *G.E.A. (Applied Electronics Group), National University of Río Cuarto* 2024 – 2026
R&D and FPGA implementation of low-latency strategies for open-switch fault detection in power semiconductor devices used in electric drives.
 - Designed and implemented HDL/RTL architecture for real-time fault detection in inverter power switches, achieving an average diagnosis time of 10% of the electrical period while using approximately 6% of LUTs and 5% of flip-flops on an Artix-7 XC7A35T at 100 MHz.
 - Validated through MATLAB simulation, Proteus, hardware-in-the-loop on FPGA, and bench testing on a 0.75 kW induction motor drive.
 - Published in *IEEE Xplore* and presented at the *8th Argentine Conference on Electronics (CAE)*, UNT, 2026.
- **Service and Network Technician**, *Cablevisión – Fibertel (Telecom)* 2013 – 2019
Diagnosis and resolution of critical faults in HFC (Hybrid Fiber-Coaxial) infrastructure; advanced technical support and network operations.
- **Self-employed Electronics Engineer**, *Electronic Solutions* 2011 – 2012
Project management and direct client support; R&D of custom electronic solutions for industrial and commercial clients.

Selected Projects

- **Amplitude-Dependent Nonlinear System Identification via Volterra Kernels and PCA** 2026
(preprint)
Development of a method for identifying and synthesizing amplitude-dependent nonlinear systems.
 - Identification via exponential swept-sine excitation at multiple amplitudes, with up to 5 Volterra kernels extracted per measurement.
 - Continuous amplitude model via PCA with shape-scale separation. Synthesis based on fixed FIR filter banks weighted by interpolated scores.
 - In validation phase. Real-time implementation on DSP/FPGA is future work.
- **Professional Ball Launcher — Trinquete (client product, IP retained)** 2026
Design and development of an end-to-end mechatronic system for professional training.
 - Distributed dual-MCU architecture (TI TMS320F280049 engine master + STM32 supervisor, linked via Modbus RTU) with multi-axis motion control (speed, spin, tilt, azimuth), sensor telemetry (current sensors, encoders), and a local HMI for operational control.
 - Full mechanical, electrical (PCB), and 3D (SolidWorks) design.
- **Time-Domain Pitch Shifting for Real-Time Audio** 2023 – 2025
Design and validation of two time-domain pitch shifting methods for embedded platforms.
 - Doppler-based pitch shifter with two crossfaded read pointers. Implemented in C on ARM Cortex-M4F (Tiva C TM4C123G) at 44.1 kHz, 2 ms latency. Works on polyphonic signals; known artifacts: transient duplication and detuning.
 - Weighted granular decimation of an oversampled buffer ($L = 4^n$). Hann window scaled to power of two for bit-shift normalization ($L = 16$, $Q = 12$). Validated in MATLAB simulation at 44.1 kHz. Preprint. Hardware implementation on STM32 BlackPill planned.

- **Digital Audio Equalizer on FPGA** 2023
5-band FIR filter bank (order 170 per band) with symmetric systolic structure and 4-level PWM-based output DAC topology.
– One DSP48 per filter. 44.1 kHz sampling, 1.5 ms latency, noise floor -60 dBVrms, ripple $< \pm 2$ dB.
– Published in *Proc. 28th AADECA Congress*, University of Palermo, Buenos Aires, 2023.
- **Multi-Channel Temperature Control for Industrial Lab Flasks** 2026
Six-channel simultaneous temperature control system for laboratory equipment.
– SSR-based burst-fire control strategy with 500 ms window and NTC sensing, implemented in C firmware on ARM Cortex-M.
– Local UI with 7-segment display and rotary encoder, designed for continuous industrial operation.

Publications and Presentations

- F. Togni *et al.*, “[Low-Latency Architecture for Open-Circuit Fault Diagnosis in Electric Drive Inverter Power Switches](#),” in *Proc. 8th Argentine Conference on Electronics (CAE)*, UNT, 2026. 2026
- F. Togni *et al.*, “[Digital Audio Equalizer Implementation using an FPGA with FIR Filter Bank Structure and 4-Level PWM-Based Output DAC Topology](#),” in *Proc. 28th AADECA Congress*, University of Palermo, Buenos Aires, 2023. 2023

Skills

- **Programming:** C, Verilog, Python (basic)
- **FPGA / HDL:** Xilinx Vivado, RTL synthesis & optimization, timing analysis, HDL/IP integration, FPGA-based DSP
- **Embedded:** ARM Cortex-M, TI C2000, STM32, PIC
- **DSP:** FIR filter banks, systolic FIR structures, Volterra kernels, PCA, PWM-based DAC, real-time audio processing
- **Hardware:** ADC/DAC, DMA, SRAM, analog signal conditioning, motor control, power electronics.
- **Communications:** I²C, I²S, SPI, UART/RS-485, USB; Modbus RTU/TCP, USB Device/Host
- **Simulation & Verification:** MATLAB/Simulink, PLECS (model-in-the-loop with embedded C), Proteus, LTspice (basic), hardware-in-the-loop on FPGA
- **Tools:** Altium Designer, SolidWorks, Code Composer Studio, Vivado
- **Industrial:** PLC/HMI (Siemens, Kinco); burst-fire control, multi-channel process control

Education

- **Electrical Engineer**, *National University of Río Cuarto*, Río Cuarto, Argentina. 2019 – 2026
– Specialization in Industrial Electronic Systems
- **Higher Technical Degree in Electromechanics**, *Technological Institute of Río Cuarto*, Argentina. 2009 – 2011
- **Mechanical Engineering (incomplete)**, *National University of Río Cuarto*, Río Cuarto, Argentina. 2006 – 2008

Courses and Certifications

- *From Algorithm to Hardware: Machine Learning in Embedded Systems*, UNMdP. 2025
- *DSP for Power Electronics Control*, U.N.R.C. 2024
- *English C1 Advanced*, [EF SET Certificate](#). 2024
- *Advanced Programming in C for PIC Microcontrollers*, Microchip Technology Inc. 2013

Languages

Spanish (native) · English (C1 Advanced)